

Writing an applied text analytics research proposal



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Acknowledgements:

Material from OCOM5300 Artificial Intelligence Project.

Textbook: C Dawson. 2015. Projects in Computing and Information Systems: A student's guide (third Edition). Pearson.

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Assessment Brief states required sections ...



Dawson (2015) outlines five features of academic computing projects.

Your project should include three or more features.

The nature of your project will influence the methods you use to tackle it.

- Review: systematic review of solutions to related tasks
- Development: software / system development
- Evaluation: comparing alternative solutions to a task
- User-based: solving a requirement for a “client/user”
- Problem Solving: alternative technique(s) to solve a known problem



Aim to impress everyone involved in your project

Who will benefit from your work?

- You (good degree?)
- Your Supervisor (Co-author of conference paper?)
- Clients and Users (Useful solution to a problem/task?)
- Evaluators and Testers (Potential users?)
- Examiners (Enjoy reading your report?)



- You may tackle existing problems with different tools or techniques, or try tools/methods in contexts where they have not been applied before.
- You may investigate a task that has been looked at before, you may come across unexpected results or different methods that have not been explored.
- You can interpret data in different ways, use them in new ways or apply them in alternative areas that have not yet been investigated.
- Originality can also be new results of your project: a new model or theory or annotated data-set

- The ‘so what?’ test
 - Ask yourself, ‘Is the topic meaningful?’ will it be of value to anybody? What contributions will it make?
 - Pursuing a meaningless project can lead to poor motivation as your project progresses.
- Explainability
 - Can you explain your project and justify it in simple terms to an average person? If so, you have a good understanding of the subject area and the topic you want to pursue.
- Contacts
 - When you identify the contacts you require for your project, are they available, accessible and willing to help? Eg users for requirement specification, evaluation

- Select recent research task, adapt to your job/company
- Select research on English, adapt to another language
- collect and label a novel specialized Corpus, for ML
- SemEval task, eg Offensive Language Identification
- PhD project outline eg <http://www.comp.leeds.ac.uk/eric>

Include use of at least two data mining and/or text analytics methods, techniques or resources introduced in this module, applied to your job/company/language.

Google Scholar



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Google Arts & Culture

Dataset Search

Google Trends

Google Scholar





Google Search Engine

Searches the internet (indexes 5% of the total internet)

Google Scholar

Searches for scholarly literature, e.g. Journals, conference papers, theses, preprints etc

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Indexes different file types, e.g. PDFs, Word, Excel, PowerPoint etc – useful for grey literature	Will give access to full text if Leeds subscribes to the content or it has been freely made available online

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Uses ranking systems (based on algorithms) to sort results. It's not clear how they are ranked.	Ranks documents based on full text availability, where it was published, who it was written by, times cited etc



In advanced search you can:

- Search in a specific site
- Search for a specific file type
- Search in a specific language
- Search in a date range
- Change the country

Use OR to search for any of your words

Use verbatim search to include all the words
(tools>verbatim)

Use “speech marks” to search for a phrase

Use minus to exclude terms (-climate)



In advanced search page you can:

- Search for title
- Search for author
- Specify a date range
- Search in a specific journal

Use OR to search for any of your terms

If you use no Boolean operator it will search for all your terms

Minus sign excludes a word, e.g. -climate

Speech marks for phrase searching

Create an alert to keep up to date with new research

Benefits of a work plan



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- Divide a large problem down into easy-to-understand tasks.
- Sharpens focus.
- Supports planning and control – includes time estimates.
- Provides progress visibility.
- Provides structure.
- Leads to better coding and documentation.

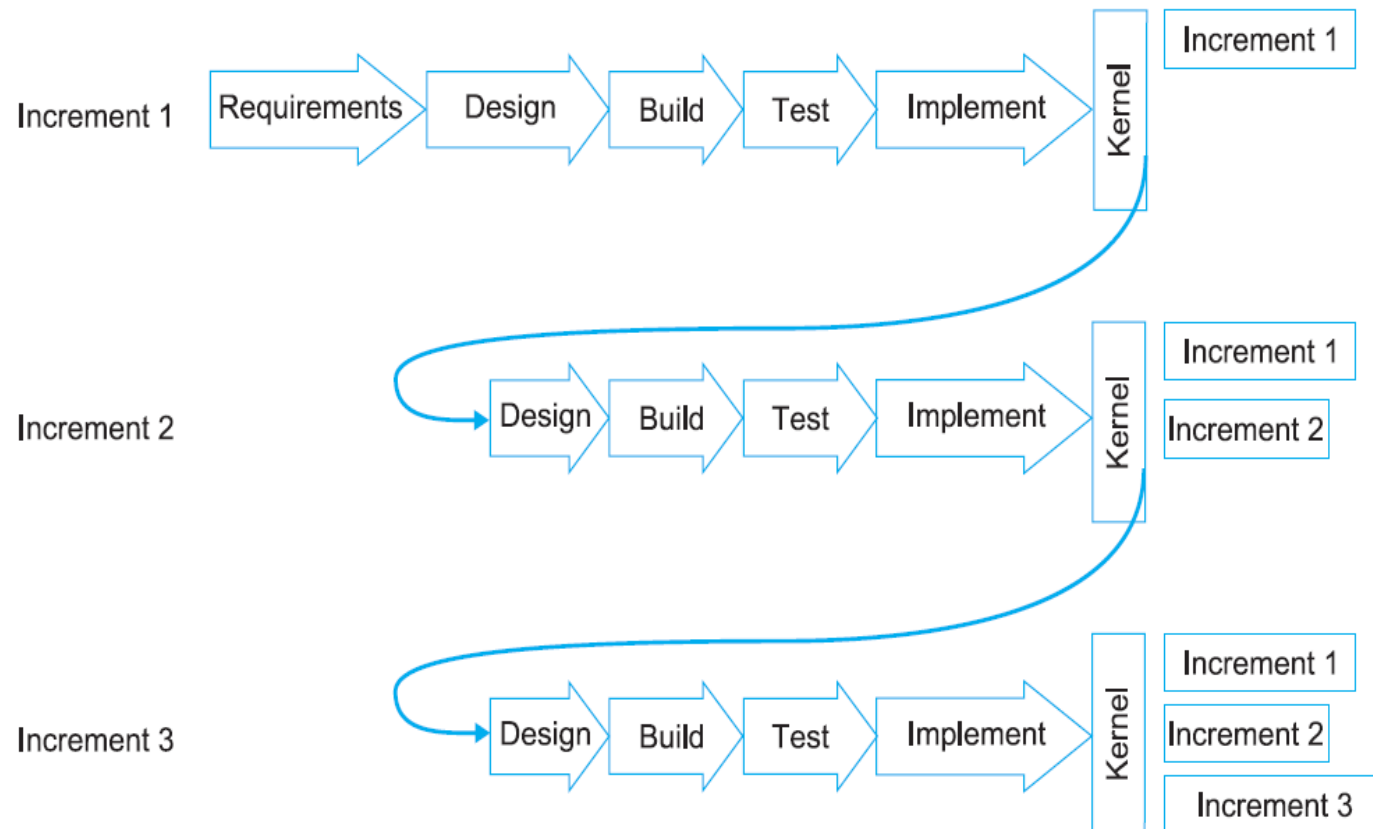

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graph TD; RA[Requirements analysis] --> S[Specification]; S --> D[Design]; D --> C[Coding]; C --> T[Testing]; T --> I[Implementation]; S --> RA; D --> RA; C --> RA; T --> RA;
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The diagram illustrates the software development lifecycle as a sequential process with feedback loops. The stages are represented by boxes in a descending staircase pattern: Requirements analysis, Specification, Design, Coding, Testing, and Implementation. Arrows show the forward flow from one stage to the next. Feedback loops are shown as curved arrows pointing from Specification, Design, Coding, and Testing back to Requirements analysis. Each feedback loop is associated with a label: 'Requirements definition' for the loop from Specification, 'Req.spec/Functional spec/System spec' for the loop from Design, 'Design spec – System design, Module design' for the loop from Coding, and '. . and Unit testing' for the loop from Testing. Additionally, there are labels for the forward flow: 'Integration and system testing' between Coding and Testing, and 'Installation and acceptance test' between Testing and Implementation.

The incremental model



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The incremental model (Dawson, 2015, p.131)



It is useful to produce a prototype in order to:

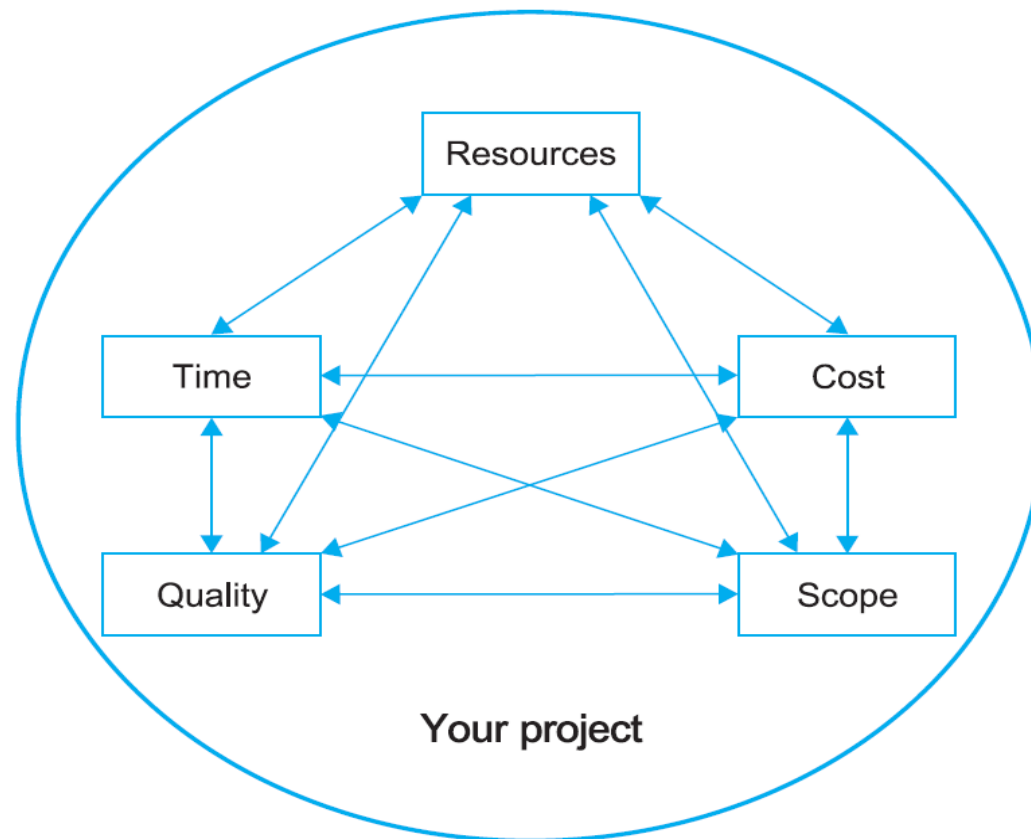
- Explore the requirements of the system with the user.
- Explore the concepts for a proposed system.
- Explore the technical feasibility of a system.

Advantages

- Helps the user to refine their requirements.
- Missunderstangs, errors and omissions in the requirements can be sorted out.
- The prototype can test the feasibility and usefulness of the product.
- Alternatives can be compared using different prototypes.

Disadvantages

- It might look messy which might give a bad impression to the user.
- If you develop it using on a different system you may not spot technical issues.
- It can take a lot of effort to develop something that you are going to eventually discard.



Balancing five project elements against one another
(Dawson, 2015, p.164)

- **Weakening – failure to work hard**
 - It can stem from a lack of motivation, losing your direction, working yourself too hard,...etc.
 - Make sure you select a topic you are really interested in.
 - Spend more time in defining and planning your project.
 - Break your work down into manageable pieces.
- **Personal problems**
 - Inform your supervisor and/or personal tutor.
- **Hardware failure**
 - Back up, back up, back up!
- **Data unavailable**
 - Set up a contingency plan. This can include changing your project's direction.

- **Discovering your research has been done before**
 - To avoid this problem, conduct a comprehensive literature review.
 - Identify 'key players' in your field and keep track of their research.
 - If another project looks very similar, identify key differences, eg in data-set, ML methods, users?
- **Results are disappointing**
 - examine why the results are “bad” - “interesting negative results”?
 - maybe conduct more experiments
 - research is not predictable
- **Algorithm/technical failure**
 - Develop a prototype first.

- Dawson, C. (2015) Projects in Computing and Information Systems: A student's guide (third Edition). Pearson

ALSO, SEE: Copilot to help draft documents ...

<https://copilot.Microsoft.com>